TAIBA N'DIAYE WIND POWER

Document Prepared by AERA GROUP

on behalf of Parc Eolien Taiba N'Diaye S.A.U

Project Title	Taiba N'Diaye Wind Power
Version	3
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Project ID	2588
Monitoring Period	01-August-2022 to 31-July-2023
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1 PROJECT DETAILS

1.1 Summary Description of the Implementation Status of the Project

- The project activity involves the installation and operation of a 158.7 MW wind farm in the commune of Taiba N'Diaye, Senegal. Parc Eolien Taiba N'Diaye (PETN) is Senegal's first large scale wind energy project and a critical part of the government's "Plan Senegal Emergent". The project has been built by Lekela and is owned and operated by Parc Eolien Taiba N'Diaye S.A.U.
- PETN is designed to generate electricity for at least 20 years through its 46 Vestas wind turbines. The generated electricity is supplied to SENELEC, the state-owned utility, under a long-term power purchase agreement (PPA). PETN commissioning since December 2019 resulted in a 15% increase in electricity generation capacity for the country, providing power for over 2 million people.
- In the absence of the project activity, electricity would have been generated by the operation of grid connected power plants, which are predominantly fossil fuel based. Therefore Taiba N'Diaye Wind Power project #2588 displaces carbon dioxide that would have been produced in the absence of the project activity by the grid connected power plants.
- The project implementation took place in three phases:
- 1. the first phase with the installation of 16 wind turbines i.e. a total power of 55.2 MW,
- 2. the second phase with the installation of 16 wind turbines with a total power of 55.2 MW,
- 3. the third tranche with the installation of 14 wind turbines with a total power of 48.3 MW.
- The main elements of the wind power plant are Vestas V126-3.45 MW turbines, pitch regulated upwind turbines with active yaw and a three blade rotor. The wind turbine utilises the OptiTip® concept and a power system based on an induction generator and full-scale converter. With these features, the wind turbine is able to operate the rotor at variable speed and thereby maintain the power output at or near rated power even in high wind speed.
- It has been estimated that approximately 450 GWh of electricity will be generated annually by the project, displacing approximately 243,695 tonnes of carbon dioxide (tCO2) per annum and over 2.4 million tCO2 over the 10-years crediting period.
- The total GHG emission reductions or removals generated in this third monitoring period is 217,484 tCO_2.
- Table 1: Technical specifications of the wind turbines:

• Rated power	• 3,450kW
• Wind class	• IEC IIIA/IEC IIB
• Standard operating temperature range	• from -20°C* to +45°C with de-rating above 30°C
• Rotor diameter	• 126m
• Frequency	• 50/60 Hz
• Converter	• full scale

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Table 2: Implementation timeline

22-28 December 2012	Public Stakeholders Meeting
21 June 2016	Environmental Impact Clearance
10 October 2016	PPA signature
09 December 2019	Phase 1 Commissioning date
06 November 2020	Phase 2 Commissioning date
30 June 2021	Phase 3 Commissioning date

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Audit Type	Period	Program	VVB Name	Number of years
Validation (Crediting period)	(09/12/2019 - 08/12/2029)	VCS	4K Earth Sciences Private Limited	10
Verification (Monitoring period 1)	(09/12/2019 - 31/07/2021)	VCS	4K Earth Sciences Private Limited	0.6
Verification (Monitoring period 2)	(01/08/2021 - 31/07/2022)	VCS	EARTHOOD SERVICES PRIVATE LIMITED	1
Verification (Monitoring period 3)	(01/08/2022 - 31/07/2023)	VCS	EARTHOOD SERVICES PRIVATE LIMITED	1
Total	-	-	-	2.6

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1.2 Sectoral Scope and Project Type

Sectoral scope: 1
Project type: Energy Industries (renewable/non-renewable sources).
The project activity is not a grouped project.

1.3 Project Proponent

Organization name	Parc Eolien Taiba N'Diaye S.A.U
Contact person	M. Alioune BA
Title	Finance Director
Address	Immeuble Riviona ,167 Avenue Lamine Gueye, Dakar
Telephone	+221 33 849 73 93
Email	Taiba@lekela.com

1.4 Other Entities Involved in the Project

Organization name	Aera Group
Role in the Project	Carbon consultant & offtaker
Contact person	M. Alexandre DUNOD
Title	Head of Certification and Deliveries
Address	28 cours Albert 1er, 75008 Paris, France
Telephone	+33 6 42 96 09 78
Email	a.dunod@aera-group.fr

Organization name	SENELEC
Role in the Project	Public utility & conceding authority
Contact person	M. Papa Toby GAYE
Title	Head of properties

Address	28 rue Vincens, BP93 Dakar, Senegal
Telephone	+221 33 839 30 30
Email	toby.gaye@senelec.sn

1.5 Project Start Date

09-December-2019, as the date of commissioning of PETN first tranche.

1.6 Project Crediting Period

The project crediting period shall be ten years, between 09-December-2019 and 08-December-2029.

1.7 Project Location

Country:	Senegal
Region:	Thies
Department:	Tivaouane
Municipality	Taïba Ndiaye
Locations under the influence of the project	Ndomor, Keur Malé, Minam, Mbayéne, Keur Birama, Keur Samba Awa, Keur Mbaye Sénoba. Taïba Mbaye, Same Ndiaye, Baïty Ndiaye, Baïty Gueye, Keur, Madiagne, Taïba Santhie, Keur Assane

The forty-six (46) wind turbines, each occupying 1,400 m2, i.e. a total land area of 7 hectares, are served by access tracks that accommodate the high-voltage electrical cables, of a length of approximately 34 km. The project site's geo-coordinates are: Latitude 15°00'36.5"N, Longitude 16°51'32.2"W.

Figure 1: Project site overview

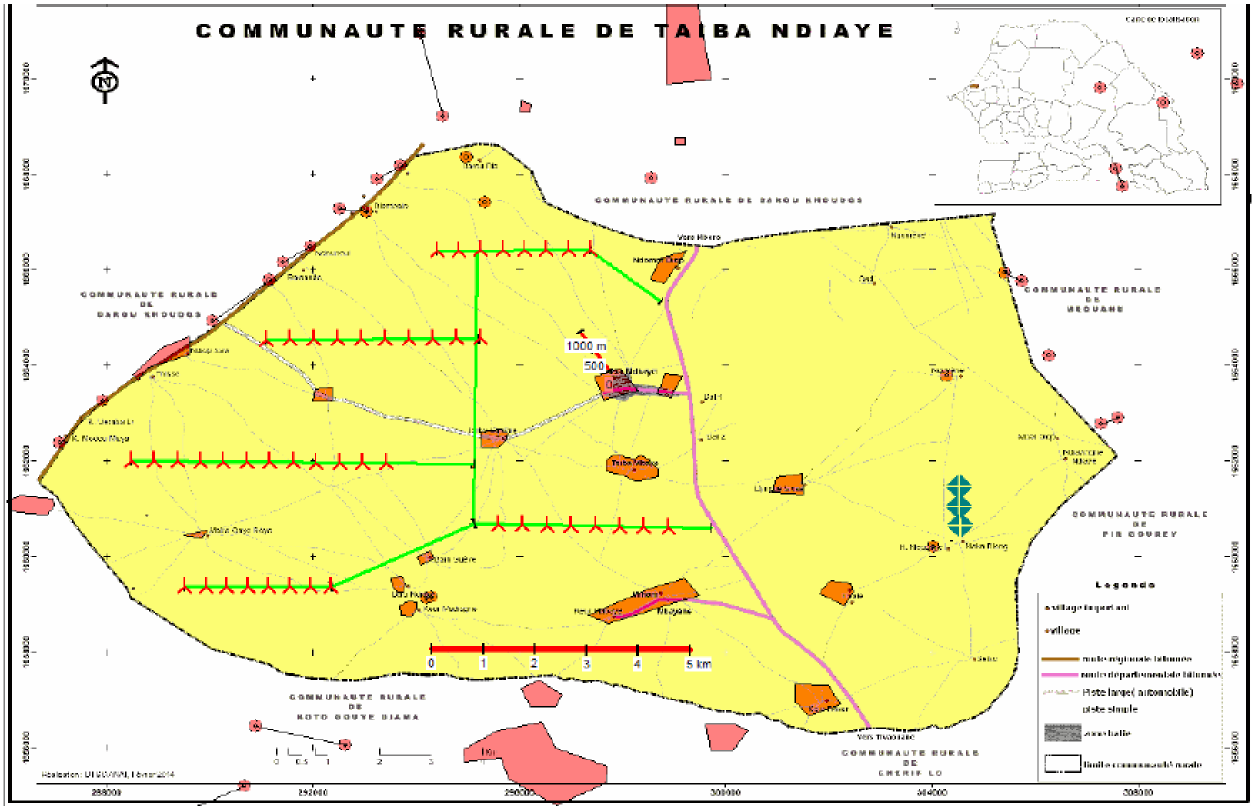


Table 3: Wind turbines coordinates

WTG label	WTG type	Hub Height [m]	"Coordinate system, UTM WGS84-28"	
			Easting [m]	Northing [m]
E1	V126-3.45 MW	117	294409	1666417
E2	V126-3.45 MW	117	294835	1666420
E3	V126-3.45 MW	117	295260	1666423
E4	V126-3.45 MW	117	295685	1666426
E5	V126-3.45 MW	117	296110	1666429
E6	V126-3.45 MW	117	296535	1666432
E7	V126-3.45 MW	117	296961	1666435
E8	V126-3.45 MW	117	297386	1666438
E9	V126-3.45 MW	117	291103	1664553
E10	V126-3.45 MW	117	291563	1664556
E11	V126-3.45 MW	117	292023	1664559
E12	V126-3.45 MW	117	292484	1664562
E13	V126-3.45 MW	117	292952	1664565
E14	V126-3.45 MW	117	293404	1664568
E15	V126-3.45 MW	117	293864	1664571
E16	V126-3.45 MW	117	294325	1664574
E17	V126-3.45 MW	117	294785	1664577
E18	V126-3.45 MW	117	295245	1664580
E19	V126-3.45 MW	117	288472	1662016
E20	V126-3.45 MW	117	288922	1662012
E21	V126-3.45 MW	117	289373	1662008
E22	V126-3.45 MW	117	289823	1662004
E23	V126-3.45 MW	117	290273	1662000
E24	V126-3.45 MW	117	290723	1661996
E25	V126-3.45 MW	117	291174	1661992
E26	V126-3.45 MW	117	291624	1661988
E27	V126-3.45 MW	117	292074	1661984
E28	V126-3.45 MW	117	292524	1661980
E29	V126-3.45 MW	117	292974	1661976
E30	V126-3.45 MW	117	293424	1661972
E31	V126-3.45 MW	117	295604	1660681
E32	V126-3.45 MW	117	296075	1660672
E33	V126-3.45 MW	117	296545	1660662
E34	V126-3.45 MW	117	297015	1660653
E35	V126-3.45 MW	117	297485	1660643
E36	V126-3.45 MW	117	297955	1660634
E37	V126-3.45 MW	117	298425	1660624
E38	V126-3.45 MW	117	298895	1660615
E39	V126-3.45 MW	117	289523	1659390
E40	V126-3.45 MW	117	289928	1659390
E41	V126-3.45 MW	117	290333	1659390
E42	V126-3.45 MW	117	290739	1659389
E43	V126-3.45 MW	117	291144	1659389
E44	V126-3.45 MW	117	291549	1659389
E45	V126-3.45 MW	117	291954	1659389
E46	V126-3.45 MW	117	292360	1659389

1.8 Title and Reference of Methodology

The approved baseline and monitoring methodology selected for to the proposed project activity is: ACM0002 version 20.0 - "Grid-connected electricity generation from renewable sources" The methodology also refers to the latest approved versions of the following tools, which are applied by the project:

- TOOL23: "Additionality of first-of-its-kind project activities" version 03.0
- TOOL07: "Tool to calculate the emission factor for an electricity system" version 7.0

- ASB0034-2021: “Grid emission factor for West African Power Pool” version 01.0

1.9 Participation under other GHG Programs

The project neither has nor intended to generate any other form of GHG related environmental credit for the GHG emission reduction or removal claimed under the VCS programme. Although it was originally registered under the CDM by its former developers[1] , it was postponed and re-designed by its new developers, for certification and verification under the VCS only.

1.10 Other Forms of Credit and Supply Chain (Scope 3) Emissions

The project does not reduce GHG emissions from activities that are included in an emissions trading program or any other mechanism that includes GHG allowance trading. The project does not involve a supply chain as project scope is limited to the reduction of fossil fuel consumption thanks to renewable energy generation.

1.11 Sustainable Development Contributions

During this monitoring report, there were no project activities implemented.

With the construction and operation of this 158.7 MW wind farm, the project ensured access to affordable, sustainable and modern energy for the commune of Taiba N’Diaye (SDG 7) and reduced GHG emissions (SDG 13).

The contribution to these SDGs allows to achieve the pillars of the national sustainable development strategy of Senegal:

- Lettre de Politique de Développement du Secteur de l'Energie

- Stratégie Nationale de Mise en Œuvre de la CCNUCC et Rapport de Contribution Prévue Déterminée au Niveau National (CPDN).

[1] As per CDM Project 5846 : Taiba N’diaye Wind Energy project, Senegal registered on 29 Feb 12, de-registered on 05 January 2022

Table 1: Sustainable Development Contributions

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
1)	7.2	7.2.1 Renewable energy share in the total final energy consumption	Renewable energy exported to the Senegalese grid from the grid-connected wind farm. (Implemented activities to increase).	379,554 MWh were delivered to the Senegalese grid during this monitoring period	1,311,821 MWh in total were produced and delivered to the grid over project lifetime.
2)	8.8	Promote sustained, inclusive, and sustainable economic growth, full and productive employment and decent work for all	Employments created and labour terms and conditions, such as job-related health and safety and training of employees. (Implemented activities to increase)	0 new jobs were created during this monitoring period	21 jobs in total were created during overall project lifetime.
3)	13.0	Tonnes of greenhouse gas emissions avoided or removed	GHG emissions reduction thanks to project (Implemented activities to increase).	The project contributed to Emission Reductions amounting to 217,484 tCO2e for this monitoring period	Total emission reductions for this project amount to 751,672 tCO2e to date.

2 SAFEGUARDS

2.1 No Net Harm

The development of the project will inevitably result in environmental and social impacts, both positive and negative

Potentially sensitive environmental component	Critical stage	Potential negative impacts	Mitigation measures
Soil and subsoil	Construction phase	• Temporary soil compaction • Soil sealing due to concreting and excavation of soil at the base of the wind turbines • Limited risk of soil and groundwater pollution by accidental infiltration of liquid pollutants (construction equipment or storage: hydrocarbons, hydraulic oils, lubricants and paints).	• Carrying out geotechnical tests in situ. • Agricultural reclamation of the road surface and the surface layer of the foundations. • Storage of dangerous liquid products (oils, fuel, etc.) during the construction site on a retention tank that can contain the entire volume of the tank. • Provision of anti-pollution intervention kits on site.
	Operating phase	• Low risk of soil and subsoil pollution: presence of oils in the wind turbines (about 1500 l/wind turbine), oils in the transformers. Appropriate retention facilities at the installations concerned should limit this risk. • Risk of pollution during maintenance and oil changes.	• Place equipment containing oils (gearbox, transformers, etc.) in a sufficiently large retention tank. • Use dry-type transformers instead of oil transformers. • Carrying out maintenance according to a well-established schedule and taking all the necessary precautions to avoid any spillage of oil or any other liquid substance dangerous for the environment. • Return used oil to approved collectors

Biotopes, fauna and flora	Construction phase	**Impact on biotopes :** • Decrease in cultivable area and potentially in initial agricultural production • dust deposit during the work ; • right of way, surface consumption; • clearing and cutting of isolated trees ; • habitat modification ; • trampling of the surrounding habitats (work, walkers) and over-frequentation of the environment; • increased risk of fire ; • contribution of invasive exogenous species ; • destruction of protected species • damage to heritage and/or keystone species and/or keystone species. • Temporary alterations to soil qualities (see impacts on soil and subsoil). • Pruning or removal of certain plantations and/or remarkable shrubs along the access roads.	• Compensate landowners according to an agreed scale • Protection of the remarkable species present in the fields and along the access roads. • avoidance of sensitive habitats and species in the project design. • Care should be taken when transporting materials. During construction work, the flora present on the site should be protected as much as possible. • application for a derogation for the destruction of a protected species must be drawn up • the marking of sensitive species before the works and the ecological follow-up of the building site;
	Operating phase	**Impacts on Birds :** • direct mortality from collisions with wind turbine blades; • disturbances and disturbances, which result in a "barrier effect", distancing and sometimes, in critical situations, a loss of habitat.	• choice of site: this is the main factor which makes it possible to reduce or eliminate the majority of the impacts on the natural environment and therefore on birds; • positioning of wind turbines to avoid barrier or funnel effects. The orientation of the wind farms parallel to the migratory axes effectively reduces the negative effects on migratory birds. • choice of the period of work and the planning of the building site according to the calendar of the species, in particular outside the periods of reproduction of the most sensitive local species, i.e. between April and September included. • Elaboration of an ornithological follow-up to

			evaluate the impacts of the wind turbines on the avifauna and, more particularly on the migratory birds. • creation of a substitute habitat with, for example, the reconstitution of a network of hedges, the creation of a land reserve away from the wind farm to maintain fallow land for wildlife • beaconing of the wind turbines is necessary to limit the impact on avifauna.
Landscape	**Construction phase:**	• direct and indirect effects on the landscape due to : • destruction of existing vegetation and opening of views; construction or widening of access roads, earthworks, tree removal, soil compaction • modification of the colour and vegetation of the site; • appearance of weeds due to the contribution of exogenous soil • partial or total artificialization of the site (roads, embankments, areas without vegetation, etc.).	• limit to the strict minimum the addition of materials, the clearing of brush and the reshaping of the track at the end of the work • maintain a bearing capacity within the runways that allows the use of motorized vehicles for • maintenance
	Operating phase	• The creation or modification of existing roads and the creation of roads for the operation and maintenance or, if necessary, for the discovery of wind turbines can have different consequences on the site: o possible over-frequentation due to the opening of new accesses or the modification of existing routes; o Conflicts of newly juxtaposed practices due to easy access to motorized vehicles; o abandonment of the site by some of its users, following the installation of the wind	• The promoter has already taken steps to ensure that the wind farm has as little impact on the landscape as possible (limiting the number of turbines, arranging the turbines in relation to the houses, installing them in a row, etc.). • Retain the assembly areas for maintenance, whereas it was previously recommended that they be removed after the installation. Integrate the head cabin in a harmonious manner. • paint the mast of the wind turbines with a non-reflective coating to prevent it from reflecting the sun "s rays. • Plantations (tree lines, etc.) or landscaping reminiscent of these characteristics will facilitate the integration of the site into the landscape.

		turbines. • Stroboscopic effect and reflection of sunlight	
Sound environment	**Construction phase:**	• Nuisance caused by the passage of construction equipment (trucks, cranes), increase in the number of peak levels per hour. • The noise generated by the construction site will be audible in the vicinity of the nearest houses, but the noise level should be below the limit values.	• Choice of the establishment and choice of the machines: research of an acoustic impact the least possible. This measure consists in choosing the establishment (number, localization of the wind turbines) and - of the machines answering the local constraints. • Carrying out the work on working days (work at weekends, at dawn and in the evening should be avoided as far as possible). • Sound power limitation (not to exceed the values considered in this study).
	Operational phase:	• Increase in background noise at the immission points (currently very quiet). • The noise of the wind farm should not exceed the limit values in the nearest residential areas, the noise is very low. • Above 8 m/s, the wind noise should mask the noise generated by the wind turbines: it can be considered that for these wind speeds, the noise will hardly be audible.	• Periodic maintenance of wind turbines to limit mechanical noise. • Carrying out an acoustic study after the park has been built. • Adaptation of the mode of production: to program an operation "with slowing down" of the wind turbines (and thus to limit the noise emissions) according to the hours of the day and the period of the week or the year.
Capacity of public facilities and infrastructure		• Passage of construction machinery and abnormal loads on the unsuitable secondary network (possible damage to roads and side tracks, noise pollution, etc.). • No impact on air traffic	• Establishment of a special convoy route in collaboration with the wind turbine construction company, the police and local authorities. • Day and night beaconing will be carried out on the wind turbines to ensure the safety of the air space.
Health and safety		• The risk of accidents (broken blades, falling masts, fire, etc.) of the rotor, environmental pollution, road transport of wind turbine components,	• Install signs warning of the dangers present on the site (falling objects, electrical risk, This signage must be placed at the entrance to the site and at each storage and lifting platform. This signage must be

		lightning).	placed at the entrance to the site and at each storage and lifting platform. • Electromagnetic interference and radiation • risk of radar interference ; • risks to aviation safety ; • risk of interference with radio waves fire hazard, etc.

2.2 Local Stakeholder Consultation

Conduct of the consultation Local Stakeholders Consultation took place from 22 to 28 December 2012, between Dakar and Thies, where the following actors were consulted:

- The National Technical Departments:
  - the Directorate of Environment and Classified Establishments ;
  - the Civil Protection Directorate;
- the technical services of the Thiès region :
  - the Regional Division of the Environment and Classified Establishments ;
  - the Regional Directorate of Rural Development ;
  - the Regional Water and Forestry Inspectorate;
  - Regional Development Agency ;
  - the Regional Planning Department ;
- the Sub-Prefect of Ouadiour ;
- the Commune of Taïba N'diaye ;
- the populations of the villages of Diambalo, Balsande, Taïba N'diaye, Taïba Mbaye, Baïty N'diaye, Baïty Guèye, Minam Diop, Mbayène, Khelkom Diop, Taïba Santhie, Maka Gaye Bèye and Ndomor Diop
- the two persons claiming ownership of the land on which the unit will be located

The technique used was interview surveys which ensured an in-depth and detailed exploration of the questions asked. These interviews were conducted on the basis of a semi-directive interview guide. In fact, this tool, through the intermediary of theme-questions, sets out the thread of the discussion between the researcher and the actors and acts as an intermediary for the exchange.

Comments and conclusion on the public consultation

The opinion shared by the different actors on the project of the implantation of a wind farm in the Commune of Taïba N'diaye is that it will allow to improve significantly the production of electricity in Senegal with a contribution of 158.7MW foreseen annually. If some inhabitants of the region are offended by the fact that their villages cannot benefit from this long-awaited service, there is an understanding of the objective of the project, which is to contribute to reducing the deficit in the supply of electricity that is plaguing many countries in the economic and social life of the country.

In terms of impacts, the issues related to the implementation of the project have not changed with regard to the problems raised. Firstly, the economic resources of the grassroots communities are affected by the annexation of land used for subsistence crops and sources of income through cash crops and harvested products that enable households to ensure their survival. From this perspective,

the discrepancy between the losses incurred and the low level of compensation paid out is the main complaint about compensation efforts. Thus, an increase in the rates applied for the care of people affected by the project, in addition to accompanying measures aimed at enabling affected people to escape the trap of impoverishment, are strong expectations expressed by the stakeholders.
Secondly, the presence of the wind turbines raises concerns about the impact on the health and safety of people in the project's area of influence. However, an improvement compared to the first project implementation plan is the increase in the distance between the wind turbines and the nearest villages by about 500 meters. Nevertheless, the head of the Regional Division of the Environment and Classified Establishments suggests increasing the distance between them and the last dwellings by at least another 500 meters, in anticipation of the growth that the human settlements are sure to experience in the medium and long term, by implanting them at least 1500 meters away.

The presence of laterite runways is another concern, with the risk of dust generation that alters the environment, the living environment and the well-being of the population. In response to this problem, it is recommended that a vegetation curtain be created along with the road axes that can help contain dust emissions. It will be a question of marking out the tracks of burial of the cables. This distance will also make it possible to guard against the risks of safety which could be amplified by a too great proximity between the dwellings and the alignments of wind turbines. From another point of view, the period of execution of the works will be determining to minimize the socio-economic impacts. The reason for this is the harvest period and the level of progress of seasonal crops.

Furthermore, the safety of people and animals may be compromised during the execution of the work. Therefore, fluorescent strips must be installed for the population. But for the animals, surveillance will be necessary, which implies at the same time an awareness of the populations.

Finally, the creation of the park raises concerns about the loss of vegetation cover and the management of hazardous waste such as PCBs, which are the main sources of impact on the natural environment. To this end, it is essentially recommended that the area be re-greened by targeting forest species and that the procedures approved for the management of dead oil during the first study be maintained, which had proposed satisfactory solutions that allowed its revalidation by the technical committee.

Grievance mechanisms (on-going communication with local stakeholders)

By communicating all contact details of the project proponent and ESIA consultant during the meetings, stakeholders were informed about the creation of the grievance mechanism through, which any complaints, suggestions, recommendations etc. could be communicated. It mainly consists of complaint forms, which are available and can be filled at the village council of Taiba Ndiaye, as well as the sub-prefecture of Meouane as well as the regional government administration of Thiès.

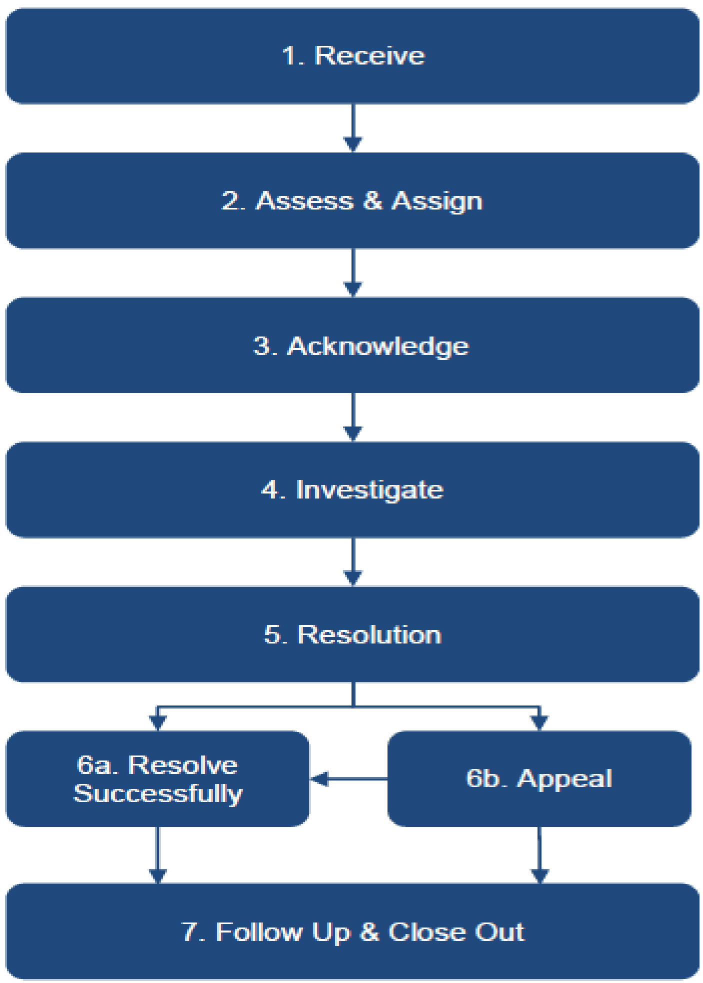
PETN will appoint an Environmental Monitoring Officer for the construction and decommissioning phases, who will, under the direction of the Project Manager, have the following main duties.

- Participate in the planning of work requiring environmental monitoring and inform the various stakeholders (contractors, construction project manager, maintenance managers and park operators) of the environmental requirements;

- Ensure compliance with the monitoring program;
- Inspect the work;
- Prepare all required reports, including monthly and annual reports required by lenders, PETN management and government authorities, where applicable.
- During the operational phase, the operations manager will be responsible for environmental monitoring.

A formalized and official PETN Grievance Mechanism Procedure (GMP) is established in order to manage grievances that may originate from the various project related activities. The GMP will address grievances that originate from the wider public (community) relating to any aspect of the project. Issues may vary from health and safety concerns to recruitment and procurement or any other issues identified by individuals who believe they have been aggrieved by the project. Figure 2 below outlines the steps that will be followed by PETN to address any grievances received.

Figure 2: PETN Grievances process



Grievances can be filed through any of the following channels:

- Via verbal communication to any PETN or affiliated site employee - Employees will be advised to direct individual grievances to a site representative designated to receive such input. Non-employees can speak to any site manager to discuss a grievance. If needed, individuals will be assisted by PETN representatives in the applicable local language.
- Via anonymous mail box on the project site and at each site entrance gate - Box will be checked daily by the PETN Construction Project Manager (“CPM”)
- Via Telephone to a dedicated hotline that will be posted and communicated through the Stakeholder Engagement Plan (SEP) outreach program
- Via Email to a dedicated address that will be posted and communicated through the SEP outreach program

During the pre-construction phase, the project has Appointed EES Sarl as Community relations manager. For submission of grievances, EES have designated Mr. Mbaye Sarr who can be reached on - Tel. 77 633 89 65 and will provide the affected persons with the relevant grievance form to register grievances.

During the third full year of operations and as documented in Annual report – January-December 2022 as well as Q1, Q2, reports, no community grievances were raised.

Grievance management mechanisms are still active and the project continues to maintain strong relations with the community.

2.3 AFOLU-Specific Safeguards

The project is not an AFOLU Project.

3 IMPLEMENTATION STATUS

3.1 Implementation Status of the Project Activity

- The project activity involves the installation and operation of a 158.7 MW wind farm in the commune of Taiba N’Diaye, Senegal. Parc Eolien Taiba N’Diaye (PETN) is Senegal’s first large scale wind energy project The project has been built by Lekela and is owned and operated by Parc Eolien Taiba N’Diaye S.A.U.
- During this monitoring period, main bearing were replaced and repaired. The wind turbines were therefore either stopped while the part was received from abroad or reduced in terms of capacity. The other turbines were stopped for maintenance operations.
- The intervention dates for main bearing issues are as follows:
- E40 is declared on the 29th of March, 2022 and it was back to operation since August,6, 2022
- E41 is declared on the 2nd of August 2022, and it was back to operation since November, 15, 2022

- The intervention dates for maintenance operations are as follows:
- E43 Curtailed at 2.2MW instead of 3.45MW since November, 8, 2022 and back to normal operation since December, 16, 2022.
- E20 Curtailed at 2.2MW instead of 3.45MW since November, 8, 2022 and back to normal operation since December, 22, 2022.
- E09 Curtailed at 2.2MW instead of 3.45MW since November, 8, 2022 and back to normal operation since January, 22, 2023.
- E02 Curtailed at 2.2MW instead of 3.45MW since November, 8, 2022 and back to operation on March, 8, 2023.
- E35 Curtailed at 2.2MW instead of 3.45MW since January, 4, 2023 and back to normal operation on April, 19, 2023..
- Apart from that, the project ran smoothly.
- Since commissioning, the project Wind Farm produced 1,311,821 MWh of clean energy which allowed to reduce 751,672 tCO_2.

3.2 Deviations

3.2.1 Methodology Deviations

- There was no methodology deviation

3.2.2 Project Description Deviations

There were no project description deviations.

3.3 Grouped Projects

The project is not a grouped project.

4 DATA AND PARAMETERS

4.1 Data and Parameters Available at Validation

Data / Parameter	$EF_{grid,CM,y}$
Data unit	tCO2/MWh
Description	Combined margin CO2 emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system”
Source of data	As per ASB0034-2021
Value applied	0.573

Justification of choice of data or description of measurement methods and procedures applied	As per the "Tool to calculate the emission factor for an electricity system" version 07.0
Purpose of Data	Calculation of baseline emissions
Comments	-

4.2 Data and Parameters Monitored

Data / Parameter	$EG_{facility,y}$
Data unit	MWh/yr
Description	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y
Source of data	Measured directly with electricity meter(s) at substation.
Description of measurement methods and procedures to be applied	Electricity outputs will be electronically stored and invoiced monthly to Senelec by PETN, after reconciliation with internal meter which readings are recorded on a record sheet by the Technical/Engineering/ Maintenance Department under the Plant Manager's authority
Frequency of monitoring/recording	Continuous measurement and at least monthly recording
Value monitored	see table below
Monitoring equipment	Bi-directional electronic electricity meters measuring the net electrical energy delivered to Senelec with the following specifications: Make: Itron Model: SL7000 Accuracy class: 0.2S (±0. 2%) Serial ID: 84406426-84406427
QA/QC procedures to be applied	Cross check of measurement results with invoiced electricity. Calibration of the meters will be carried out after each deviation of more than +- 0.5% between the 2 meters as per manufacturer & PPA specifications.
Purpose of the data	Calculation of baseline emissions
Calculation method	Electronic recording

Value monitored:

Year	2022 (from 01/08/2022)	2023 (until 31/07/2023)	Total
$EG_{facility,y}$	123,245	256,309	379,554

Comments	Last EnergiExpress inspection/testing was undertaken on 07/12/2022 as per report, valid until 06/12/2023. Previous inspection/testing was done on 09/12/2021, and valid until 09/12/2022.

4.3 Monitoring Plan

The proposed project activity monitoring plan complies with the methodology ACM0002 - Consolidated baseline methodology for grid-connected electricity generation from renewable sources, whereby it is stated that:

"All data collected as part of monitoring should be archived electronically and be kept at least for 2 years after the end of the last crediting period".

Therefore, the quantity of net electricity generation supplied by the project plant to the grid will be reliably monitored through calibrated electricity meters and cross-checked with sales records as per internal monitoring practices.

Monitoring team and training

Data collection, consolidation and results analysis will be undertaken by a dedicated team adequately trained, well aware of VCS requirements. This team will not have any hierarchical relationships or dependence links with all entities involved to measure net electricity supplied to the grid and to assure the correct operation and maintenance of the measuring equipment. This independence shall guarantee the integrity of the work that will be done.

Quality management procedures

The implementation of a system of supervision and control that will transfer real-time all monitoring information for off-site recording will allow the operator to access and retrieve safely all Project data regardless of eventual incidents.

Meter measurements by Senelec main/back-up shall not exceed +- 0.5% difference, beyond which dysfunctional meter will in all cases need to be identified by Senelec and PETN, adjusted or replaced within 48 hours in accordance with manufacturer guidelines. After each deviation of more than +- 0.5%, a test and calibration of the meters will be carried out for each meter, certified by a third party.

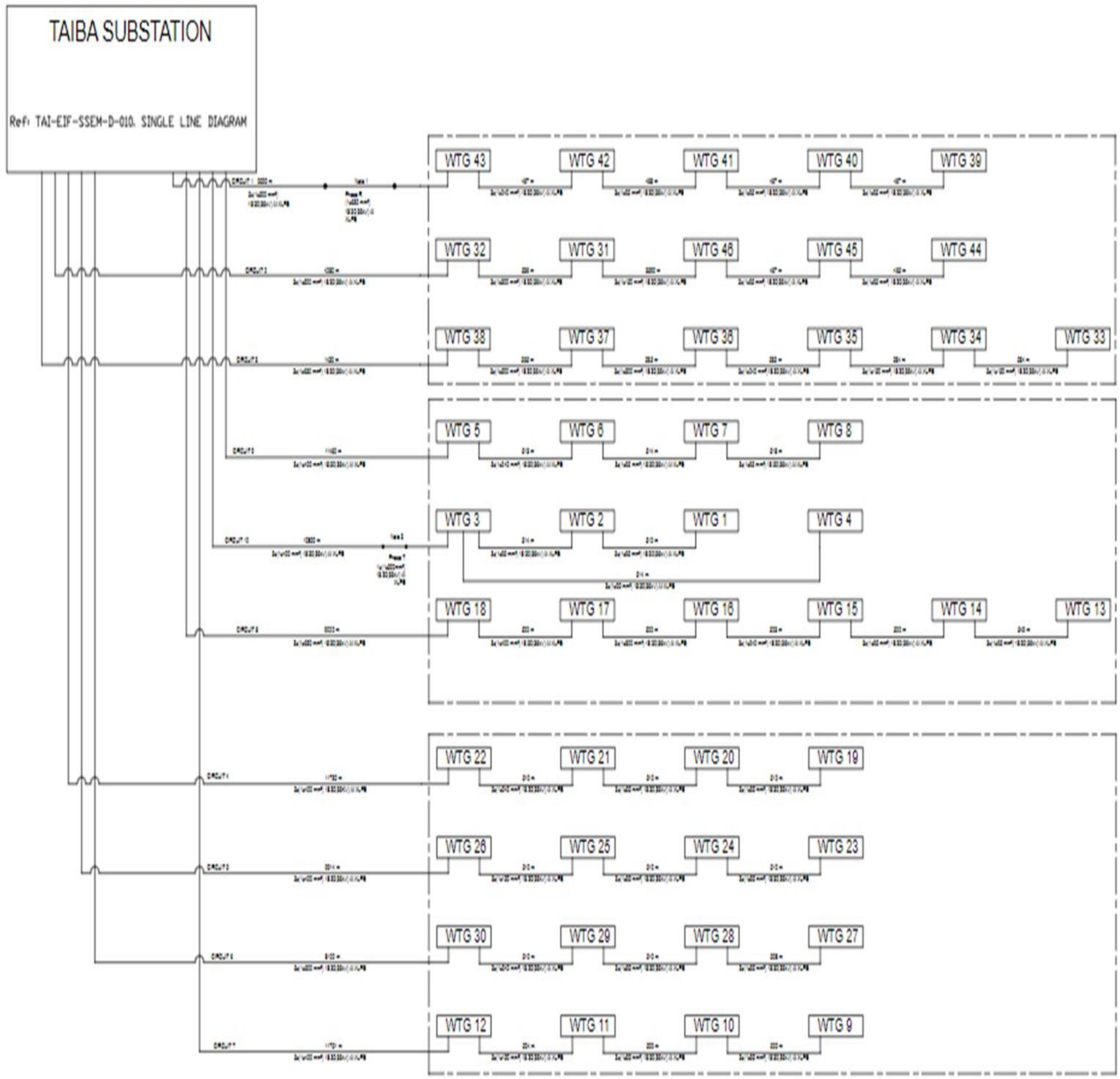
Non-conformance data during uncalibrated periods will systematically be discounted by the corresponding error margin.

Table 4: Characteristics of the meters:

Make	Itron
Model	SL7000
Quantity	2
#ID1 **#ID2**	84406426 84406427
Accuracy	0.2s

Calibration date	07/12/2022 valid until 06/12/2023
Error identified for main meter during last calibration	-0.06%
Error identified for back-up meter during last calibration	-0.06%
Calibration certificate number	0072-EE

Figure 3: Line Diagram



5 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

The monitoring period for which GHG emission reductions were achieved spans 01/08/2022 to 31/07/2023. Baseline emissions are firstly calculated based on the difference between the total quantities imported and the total quantities produced that were recorded by the main meter and the back-up meter and which are extracted from the monthly reconciliations, once total production and consumption is available for each month, baseline emissions are calculated based on §4.1 methodological approach as the product of (i) the quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the project activity in year y ($EG_{facility,y}$ = Exports – Imports, in MWh/yr) and (ii) the combined margin CO_2 emission factor for grid connected power generation in year y ($EF_{grid,CM,y}$):

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$$

Where:

BE_y	Baseline emissions in year y (t CO_2/yr)
$EG_{PJ,y}$	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the project activity in year y (MWh/yr)
$EF_{grid,CM,y}$	Combined margin CO_2 emission factor for grid connected power generation in year y calculated using the latest version of the "Tool to calculate the emission factor for an electricity system" (t CO_2/MWh)

$EG_{PJ,y} = EG_{facility,y}$ i.e. the quantity of net electricity generation supplied by the project plant to the grid, as monitored and displayed below on a monthly basis.

$EF_{grid,CM,y}$ = 0.573 tCO_2/MWh, as the Project Proponent has applied approved standardized baseline- ASB0034-2021 Grid emission factor for West African Power v1.0.

Year	BE_y	$EF_{grid,CM,y}$	$EG_{facility,y}$
	t CO_2	tCO_2/MWh	MWh
2022 (from 01/08/2022 until 31/12/2022)	70,619	0.573	123,245
2023 (from 01/01/2023 until /31/07/2023)	146,865	0.573	256,309
Total	217,484		379,554

5.2 Project Emissions

No project emissions are considered as the project activity only involves renewable electricity generation from the wind power plant. Hence according to ACM0002 guidelines PEy = 0.

5.3 Leakage

As stated in the applicable methodology, no leakage emissions are considered, therefore LEy = 0.

5.4 Net GHG Emission Reductions and Removals

Year	Baseline emissions or removals (tCO_2e)	Project emissions or removals (tCO_2e)	Leakage emissions (tCO_2e)	Net GHG emission reductions or removals (tCO_2e)
01-August-2022– 31-December-2022	70,619	0	0	70,619
01-January-2023—31-July-2023	146,865	0	0	146,865
Total	217,484	0	0	217,484

Ex-ante emissions reductions /removals	Achieved emissions reductions /removals	Percent difference	Justification for the difference
257,735	217,484	-16%	Due to main bearing issues and maintenance operations, as described in section 3.1.